

Niklas Rieken

Curriculum Vitae

Komphausbadstraße 25

52062 Aachen

Germany

+49 1523 8736069

rieken@oms.rwth-aachen.de

ttnick.github.io

ttnick

Personal Data

Date of Birth November 9, 1992
Place of Birth Krefeld, Germany

Education

May 2019 **Master of Science, RWTH Aachen, Computer Science**
Thesis: Network Pricing Games with Atomic Splittable Followers
Final Grade: 1.4
March 2017 **Bachelor of Science, RWTH Aachen, Computer Science**
Thesis: Alphabets and Machines as Structures and Their Decision Problems
Final Grade: 2.5
May 2012 **Abitur (German A-Levels), Michael-Ende-Gymnasium, Tönisvorst**
Advanced Courses: Math and Physics
Final Grade: 2.6

Coursework

- Complexity Theory
- Recursion Theory
- Mathematical Logic I + II
- Logic and Games
- Applied Automata Theory
- Infinite Computations
- Convex Optimization
- Operations Research I + II
- Optimization with Modelling Languages
- Methods and Applications of Optimization
- Discrete Differential Geometry
- Satisfiability Checking

Work Experience

2019–today **PhD Student, Chair of Management Science, RWTH Aachen**
Researcher in Mathematical Optimization and Algorithmic Game Theory.
Matchings, matroids, mechanism design, teaching courses in mathematical economics.
2017–2019 **Research Assistant, Chair of Operations Research, RWTH Aachen**
Developer for GCG (*generic column generation*, MIP solver).
Performance tests, extracting statistics, and collecting information on vast instance sets.
2014–2019 **Teaching Assistant, Chair of Computer Science 7 / TCS, RWTH Aachen**
TA for undergraduate courses in theoretical computer science.
Teaching exercise groups, grading assignments and exams.

Skills

Operating Systems **GNU/Linux**
Programming **Python, Bash, C/C++**
Applications **LaTeX(+TikZ), Gurobi**

Languages

German native
English proficient
Spanish basics

Professional Interests

- Mathematical Optimization
- Theory of Formal Languages
- Algorithmic Game Theory

Personal Interests

- Ultimate and Table Tennis
- Guitar and Ukulele
- Reading, Movies, and Music
- Sports Broadcasts